Anthony Schurle

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Education

Rice University — Houston, TX
Bachelor of Science in Computer Science — *August 2024 – May 2028*

- GPA: 3.7
- Trustee Distinguished Academic Scholarship

Experience

Pool Lifeguard — May 2024 – August 2024
Greater Houston Pool Management — *Houston, TX*

- Recognized as **Lifeguard of the Month** for exceptional vigilance, professionalism, and teamwork
- Oversaw safety of 100+ daily patrons by enforcing rules and executing proactive safety interventions
- Responded to and helped resolve 5 safety incidents through effective coordination during real-time emergencies
- Maintained pool health by regularly monitoring and adjusting pH and chlorine levels

Varsity Cheerleading Team Member — August 2024 – Present
Rice University Athletics — *Houston, TX*

- Balanced 15+ hours of weekly training, performances, and athletic commitments with full-time coursework
- Trained and performed advanced tumbling, stunting, and choreography to support athletic teams and engage spectators at home and away games
- Represented the university at donor and alumni events

Projects

Personal Portfolio | *Python, TypeScript, SASS, HTML5, PostgreSQL* — June 2025 – Present

- Built a full-stack portfolio and project showcase with a **FastAPI REST API** backend and responsive TypeScript frontend, hosted at www.anthonyschurle.com
- Containerized backend and database services using **Docker Compose** with persistent volumes; achieved deployment time under 10 minutes
- Configured **Nginx** as a reverse proxy with HTTPS (**Certbot**) and managed DNS through **Cloudflare**; achieved A+ SSL Labs score
- Designed normalized **PostgreSQL** schemas for projects, skills, and contact forms with **SQLAlchemy** + **Pydantic** validation

SignalSiege (Go-Inspired Game) | *Python, TypeScript, SASS, HTML5, PostgreSQL* — May 2025 – Present

- Developed a fully containerized browser strategy game inspired by Go, with custom grid-based logic and a graph-modeled board using **Python** and **FastAPI REST API**; project totals 3.4k LOC evenly split between backend and frontend
- Engineered a turn-based AI system with difficulty scaling, leveraging game-state deep copies, BFS traversal, and statistical heuristics to simulate optimal router placements; AI moves computed under 100ms on a 10x10 grid
- Implemented router placement rules enforcing territory control, liberty checks, and capture conditions via **graph connectivity algorithms**
- Built full serialization and deserialization of the game board, enabling AI self-play, persistent saves, and planned replay/multiplayer features
- Implemented secure login and registration endpoints using **JWTs** stored in **HTTPOnly cookies** for client-side session persistence; planned support for session-based **OAuth2**
- Persisted all gameplay stats (per difficulty), user preferences, and game states in normalized **PostgreSQL** schemas; entire project containerized with **Docker Compose** using separate app and database services
- Planned roadmap features: multiplayer engine using **WebSockets** and reinforcement learning (RL)-based adaptive AI agents

Technical Skills

Languages: Python, SQL (PostgreSQL), TypeScript (JavaScript), HTML5, CSS, R, SASS
Frameworks: FastAPI, Gunicorn, Uvicorn
Developer Tools: Git, Docker, Docker Compose, VS Code, GitHub
Libraries: SQLAlchemy, pandas, numpy
DevOps & Deployment: Nginx, Certbot (Let's Encrypt), Cloudflare DNS, Linux Shell (Ubuntu), VPS hosting